AUS RACING NEWSLETTER

MAY 2025

UNDERSTANDING FASTENERS

LEARN ABOUT THE COMPLEXITY OF A SIMPLE BOLT AND MORE

AUS RACING FSUK-CC SUBMISSION

CAR ENGINEERING REPORT HAS BEEN SUBMITTED. LEARN ABOUT WHAT'S NEXT?

WHAT CAN YOU EXPECT?

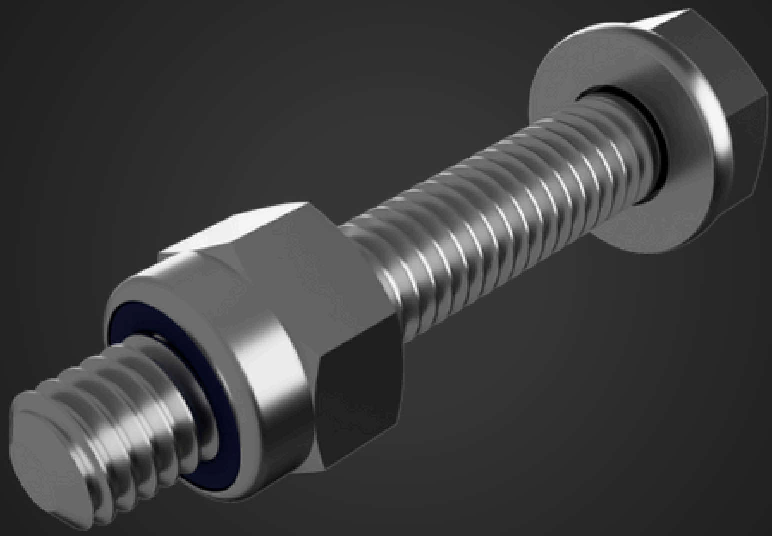
WHAT IS AUS RACING

AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

THE FASTENERS



The car, unfortunately, is not made of one singular part. Thus, fasteners are needed to hold all the parts together.

Fasteners rely almost completely on friction to ensure they do not loosen. However, over time, vibrations can loosen them and consequently weaken critical joints.

Vibrations create a sliding movement which causes the effective coefficient of friction to reduce, resulting in loosening of the fastener.

Reducing the effect of this friction reduction can be done by techniques such as preloading the fastener, using a more ductile bolt etc.

THE

FINAL CAR

AUS Racing is proud to announce that we have officially submitted our engineering design for FSUK-CC. This marks a huge milestone for the team, demonstrating the team's accomplishment. Our design is a testament to the team's diligence and determination, and we hope that we continue down this path of development. Moving forward, we will be re-iterating our design to enhance its value and optimize its efficiency. We hope our team can keep up its spirit in the last leg of this race!

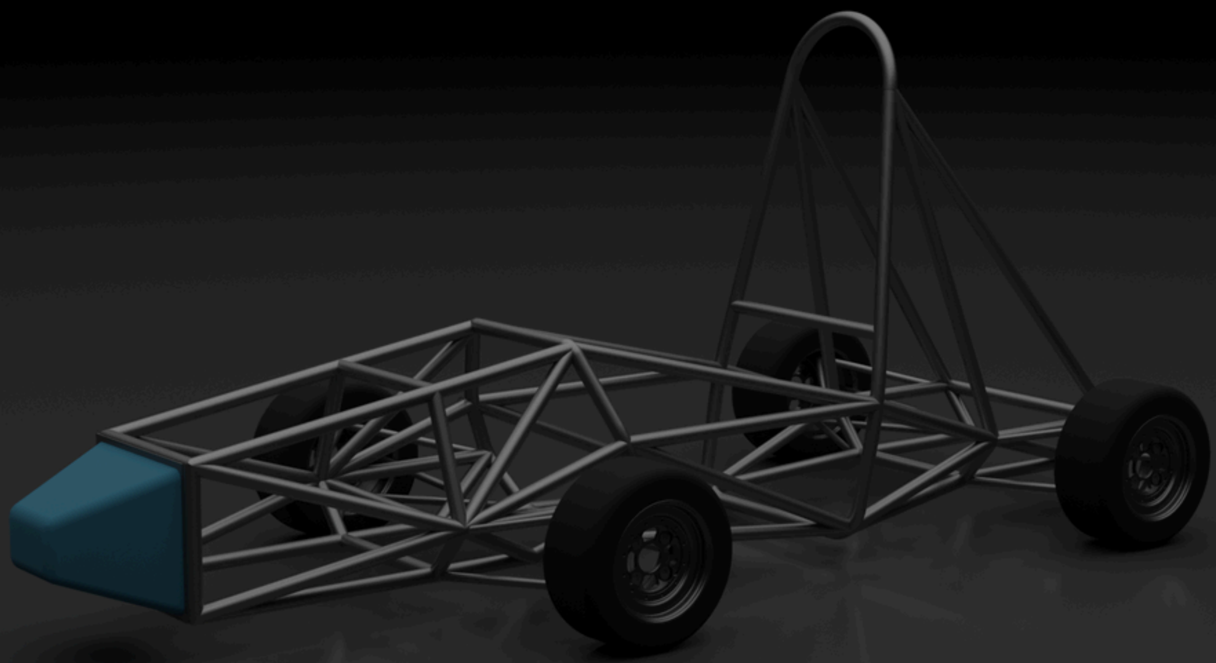
HOW YOU CAN SUPPORT US

What if you had the front row seat to engineering pioneers at work?

What if you were the reason these budding innovators were changing the future?

What if your institution could get their message to the rest of the world through Formula Student?

If your company's answer to any of these questions is yes, contact oaaa@aus.edu to learn more about getting involved with the AUS Racing team.



AUS

American University of Sharjah

GOLD

Ansys

BRONZE

AUTOMECH GROUP

IPG